

🎬 Define Function F(x)

- Input: A number x
- Output: Return the value of the expression $x^2 - 4x - 10$

🎬 Define Procedure for Regula Falsi Method

- Input: a, b (initial guesses), $noit$ (maximum number of iterations), acc (acceptable accuracy)
- **Check if the root is bracketed**
 - If $F(a) * F(b)$ is positive, print "Roots are not bracketed by a and b " and stop the program.
- **Initialize variables**
 - Set x_m (initial guess for root using Regula Falsi formula)
 - Set iteration counter n to 0.
- **Loop until root is found or max iterations reached**
 - Repeat:
 - Compute x_m (the approximate root) using the formula:
$$x_m = a - (F(a) * (b - a)) / (F(b) - F(a))$$
 - If $|F(x_m)|$ is less than or equal to the acceptable accuracy (acc), stop the loop.
 - Check the sign of $F(x_m) * F(a)$:
 - If it's negative, the root lies between a and x_m , so set $b = x_m$.
 - Otherwise, set $a = x_m$.
 - Increment the iteration counter n .
 - Continue the loop until either $|F(x_m)|$ is less than acc or the number of iterations reaches the limit $noit$.
- **Print the result**
 - Output the root x_m and the number of iterations n .

🎬 Main Program

- Set initial guesses $a = 4.0, b = 6.0$
- Set accuracy $acc = 0.00001$
- Set maximum iterations $noit = 30$
- Call the Regula Falsi procedure with these inputs.